Asita Thangavel

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Objective

Driven and adaptable undergraduate student seeking opportunities to contribute, learn, and grow in a professional environment. Passionate about leveraging technology to solve real-world problems while continuously enhancing technical expertise, analytical thinking, and software development skills.

Academic Credentials

Dr. N.G.P Institute of Technology 2023 – 2027
B.Tech Artificial Intelligence and Data Science — CGPA: 9.1 (till Semester 7)
Honours in Blockchain Technology — CGPA: 8.5

Vidyaa Vikas International School 2022 – 2023
Senior Secondary Examination (Class XII) — Percentage: 93%

Vidyaa Vikas International School 2020 – 2021
Secondary School Examination (Class X) — Percentage: 94.8%

Projects

Gesture-Based Mathematical Expression Recognition System | *Python, OpenCV, MediaPipe, Groq API, SymPy*
- Developed a real-time AI system enabling users to write equations in the air using hand gestures
- Implemented gesture recognition and hand landmark tracking using MediaPipe and OpenCV
- Integrated Llama 4 Vision through Groq API for handwritten mathematical expression recognition
- Built symbolic equation solving pipeline using SymPy

Tropical Cloud Cluster Detection | *PyTorch, U-Net, MobileNetV2*
- Developed a deep learning system for detecting Tropical Cloud Clusters from satellite infrared data
- Processed and analyzed INSAT-3D infrared brightness temperature data for cloud cluster identification
- Implemented U-Net architecture with MobileNetV2 backbone for cloud segmentation
- Built preprocessing and training pipelines for spatiotemporal satellite datasets

Transformer-Based Language Model | *PyTorch*
- Built a Transformer model from scratch implementing multi-head self-attention and positional encoding
- Configured architecture with approximately 90M trainable parameters
- Developed custom tokenization and training pipelines for sequence modeling

Visitor Management System | *React, Django, PostgreSQL*
- Built a full-stack visitor management application with approval and scheduling workflows
- Developed admin dashboards for submitted, approved, rejected, and rescheduled requests

Smart Attendance System | *Python, OpenCV*
- Developed an AI-powered attendance system using face recognition from live video streams
- Improved attendance accuracy and reduced manual tracking overhead

Internship

AI Engineer - Intern 18 June 2026 - Present
KESS, Singapore
- Worked on a RAG-based AI application for intelligent information retrieval and response generation
- Assisted in developing and validating multilingual generation capabilities for AI-driven applications
- Worked on a Machine Learning-based Fault Detection and Diagnostics (FDD) system for building equipment monitoring

Full Stack Development Intern June 2025
Pineshere Solutions, Coimbatore, India
- Worked on full-stack web development using modern frontend and backend technologies
- Contributed to application development, debugging, and deployment workflows

Technical Skills

Programming Languages: Python
AI / Machine Learning: PyTorch, OpenCV, LangChain, Generative AI, RAG, LLMs, Fine-Tuning
Web Development: React, FastAPI, Django
Databases: MySQL, PostgreSQL, ChromaDB
Tools: Git, Google Colab, Canva, Figma, Postman
Core Concepts: Operating Systems, Database Management Systems, Computer Networks

Certifications

- IBM Coursera – Machine Learning with Python
- HackerRank – Python Basics
- FreeCodeCamp – Responsive Web Design
- NPTEL Swayam – Human Computer Interaction (Elite + Gold, 94%)

Achievements & Participation

Achievements

- Top 100 Team - ISRO BAH 2025 Hackathon
- Second Rank Holder - Department of Artificial Intelligence and Data Science
- Consistent Academic Performance - CGPA: 9.2
- 1st Prize - Pitch Your AI event, GIZMO'24 National Level Technical Symposium (22–23 April 2024), Erode Sengunthar Engineering College

Technical Participation

- Participated in GIZMO'24 National Level Technical Symposium (22–23 April 2024) - Technical/Non-Technical Events
- Presented paper titled "Optimizing Energy Using User-Friendly Interface" at IGNIS'23, Bannari Amman Institute of Technology (5–6 December 2023)
- Participated in National Level Technical Symposium at PSG Institute of Technology and Applied Research
- Attended "LangChain AI Workshop" during YUKTA:24, PSG Institute of Technology and Applied Research (15–16 March 2024)
- Participated in multiple Hackathons, Codathons, Paper Presentations, AI/ML Workshops, and Innovation Events
- Participated in Paper/Poster Presentation at AAVISHKAR 2K24 National Level Technical Symposium, Coimbatore Institute of Technology (23 March 2024)

Leadership & Extra-Curricular Activities

Student Development Council 2025 – 2026
President

- Led and coordinated student activities, technical events, and organizational initiatives
- Managed event planning, team coordination, and student engagement activities

National Cadet Corps (NCC) 2023 – 2026
Company Quartermaster Sergeant (CQMS)

- B and C Certificate Holder with A Grade
- Participated in leadership training, drills, camps, and organizational activities

Robotics Club 2024 – 2025
Co-Chair Person

- Organized technical workshops and collaborative project activities
- Coordinated club operations and student participation in innovation-focused events